WasteRisk 360: An Intelligent Waste Risk Assessment, Prioritisation and Mitigation System

Team Gemba Guardians · Tatusta Hackathon (ISTE TCET) · Selected Domain: Sustainability – Waste Management. Items in [square brackets] must be filled in with your real information before submission.

1. Problem Statement

Educational institutions, residential communities and other organisations generate multiple categories of waste. Inadequate segregation, missed collection, overflowing bins, unsafe storage and improper handling of potentially hazardous waste can create environmental, operational and health-related risks.

However, waste-management decisions are often based on periodic inspections and reactive interventions rather than systematic risk assessment.

There is a need for an accessible, data-driven decision-support system that enables institutions to identify waste-management risks, quantify their relative priority, recommend mitigation measures and assess the potential effectiveness of corrective actions.

Our pilot focus:

[User, e.g. the housekeeping supervisor of Hostel A / the manager of the college canteen / the committee of Society B] has to keep four waste streams separate: wet, dry, sanitary and special care. India's Solid Waste Management Rules, 2026 (in force from 1 April 2026) require segregation at source into these four streams. [Verify this against the official notification before submission.]

At [site], failures such as [failure 1, e.g. wet waste in dry bins], [failure 2] and [failure 3] occur, but decisions about what to fix first rely on occasional inspections and personal judgement. Each failure has a different consequence, for example [kg of recyclable waste contaminated], [days of disruption] or [a compliance gap], and a limited budget cannot fix everything at once.

[One verified fact about segregation or waste handling in India, with its source. NEEDS SOURCE.]

The gap: there is no simple, evidence-based way for a small institution to (1) measure how often each failure happens, (2) rank failures by risk, (3) assign corrective actions, and (4) check whether those actions worked.

2. Proposed Solution

WasteRisk 360 is an interactive waste-management risk intelligence platform inspired by the ISO 31000 risk-management process and incorporating Failure Mode and Effects Analysis (FMEA). The application enables users to enter waste-management audit observations, assign Severity, Occurrence and Detection scores, and automatically calculate Risk Priority Numbers (RPN). It generates dynamic risk rankings, visual risk heatmaps and prioritised mitigation recommendations. Its scenario simulation module enables users to change risk parameters and compare risk scores before and after proposed mitigation interventions.

What makes it evidence-based: it links **measured audit evidence** to **risk scoring** and **mitigation tracking**, so Occurrence comes from counted observations and not only from opinion.

It has four connected modules:

1. **Evidence module.** Users record audit observations (one item placed in one bin is one event). The app calculates defect rates, a ranked Pareto chart of failures, and a matrix of expected versus actual stream.
2. **Risk module.** Each failure mode (8 to 12 in total) is scored for Severity, Occurrence and Detection. Occurrence is calculated from the audit data using written bands, so the score responds to real observations. RPN = S × O × D.
3. **Action module.** Each risk gets a mitigation, an owner, a deadline and a status.
4. **Scenario module.** Users enter an assumed reduction for a failure and see projected scores. A follow-up audit is recorded separately, so projected and observed results are never mixed.

3. Key Features

1. Audit entry form and CSV import, with validation and duplicate checks.
2. Defect rate with sample size, and the number of events behind every figure.
3. Pareto chart of failure modes with cumulative percentage.
4. Expected-versus-actual stream matrix.

5. FMEA register with Occurrence calculated from audit counts.
6. Risk ranking and a severity-S versus occurrence-O heatmap (bubble size = RPN).
7. High-severity review flag, independent of RPN.
8. Mitigation actions with owner, deadline, status and an overdue flag.
9. Before and after simulator: baseline, projected and observed follow-up shown separately.
10. Sensitivity check: each score is varied by ±1 to show whether the top three risks change.
11. Evidence page with definitions, sources, assumptions, limitations, tools used, and CSV export.

4. Methodology

4.1 ISO 31000-inspired process mapped to the app

Step	In the app
Establish context	Site, user, waste streams, definitions, scoring scales
Risk identification	Failure-mode list built from observation and the rules text
Risk analysis	Defect counts, Pareto, S/O/D scoring, RPN
Risk evaluation	Ranking, heatmap, high-severity flag
Risk treatment	Mitigation actions with owners and deadlines
Monitoring and review	Follow-up audit and observed versus projected comparison

4.2 Definitions

- **Event:** one observed disposal of one item into one bin.
- **Opportunity:** one chance to place that item in the correct stream (one per event).
- **Defect:** an event where the actual stream differs from the expected stream.
- **Uncertain event:** an item whose correct stream could not be decided. Uncertain events are excluded from the rates and reported as a separate count.

4.3 Formulas

- **Defect rate** = defects ÷ events.
- **Defects per million opportunities (DPMO)** = defect rate × 1,000,000, shown as a secondary figure only. With small samples it is a rough indicator and is not presented as a sigma level.
- **Occurrence (O)** is taken from the share of events in which the failure occurred, using these bands: no events = 1; under 1% = 2; 1 to 3% = 4; 3 to 5% = 6; 5 to 8% = 8; 8% or more = 10. [Review these bands with your mentor and adjust.]
- **Severity (S)** is scored 1 to 10 on a written scale linked to the consequence unit, for example 1 to 3 = minor, no real contamination; 4 to 6 = contamination of a stream or short disruption; 7 to 8 = hygiene or compliance exposure; 9 to 10 = potential injury or hazardous exposure. [Finalise with the site supervisor.]
- **Detection (D)** is scored 1 to 10, where a higher number means the failure is harder to catch before its consequence occurs, based on existing checks. [Define the scale.]
- **RPN** = S × O × D.
- **High-severity flag:** any failure with S at or above [8] is reviewed regardless of RPN. This is the team's own escalation rule. It is not an industry standard.

4.4 What-if rule

For a chosen failure with an assumed reduction r:

- projected events removed = count × r;
- new defect rate = (total defects − removed) ÷ events;
- Occurrence is recalculated from the reduced count;
- Severity stays the same unless the consequence itself changes;
- Detection changes only if the action improves detection.

This result is labelled **Projection only**. Only a follow-up audit using the same definitions can be labelled **Observed**.

4.5 Known limitations of the method

RPN treats S, O and D as equal and can hide a high-severity, low-frequency failure, which is why the separate high-severity flag exists. Scales are partly judgement-based, so they are written down and tested with the sensitivity check.

About the word “Intelligent”: the intelligence of the system is rule-based and automated: scoring, ranking, recommendation and scenario comparison. It does not use machine learning or AI models for its calculations.

5. Data and Evidence Strategy

- **Real pilot:** observation-only audits at [site], with permission from [manager/warden/committee], over [period]. Target: at least [N] events at baseline and [N] at follow-up. [Fill in with your actual numbers.]
- **Safety and privacy:** observers do not open bags or handle sanitary or special-care waste, and no faces or personal details are recorded.
- **Synthetic data:** a clearly labelled sample dataset is included so the application can be demonstrated offline and tested. A visible banner marks any synthetic data on screen.
- **Expert review:** the failure list and the S, O and D scales were reviewed by [name, role, e.g. a certified Six Sigma Black Belt]. [Confirm this before stating it.]
- **Reporting rule:** the documentation states which figures are real and which are synthetic, and gives the sample size next to every result.

6. Technology Stack and Architecture

Stack: Python, Streamlit (multipage), pandas, NumPy, Plotly, SQLite or CSV, pytest, GitHub. The application runs fully offline with no external APIs. [List every AI tool, API and library used in the Evidence page, as the hackathon rules require.]

Pipeline:

1. Input: audit form or CSV import.
2. Validation: required columns, valid dates and streams, duplicates.
3. Evidence calculations: defect rate, Pareto, matrix.
4. Risk calculations: Occurrence from counts, RPN, flags, heatmap.
5. Actions: owner, deadline, status.
6. Scenario: projected result, then observed follow-up.
7. Output: dashboard, risk register and export.

Code layout: calculations live in a `core/` package (data, metrics, FMEA) and are covered by automated tests. Pages only display the results.

Pages: Home, Audit Entry, Dashboard, Risk Register, Actions and What-if, Evidence and Export.

7. MVP Scope

Included: all features in section 3.

Excluded from the prototype: login and user roles, live sensors, cloud database, image recognition, prediction models and integration with government systems.

8. Expected Impact (measurable)

WasteRisk 360 aims to help educational institutions and residential communities:

- Identify high-priority waste-management risks.
- Improve waste segregation and handling practices.
- Prioritise corrective and preventive actions.
- Allocate resources more effectively.
- Monitor the potential effectiveness of mitigation interventions.

These aims are supported by the measured figures below, not by assumptions.

Impact will be shown with measured figures, not assumed ones:

- baseline defect rate compared with the follow-up defect rate at [site];
- ranked list of failures and the share of defects covered by the top [two] failures;
- RPN before and after a mitigation, with the projected and the observed values shown separately;
- time taken to produce a ranked risk report from audit data, compared with doing it by hand [measure this if you claim it].

9. Limitations and Ethics

- Results come from a short, small-sample pilot at one site and are not statistically generalisable.
- S and D scores involve judgement. The scales and the sensitivity check make this visible.
- The app is a decision-support aid and does not predict real environmental or health outcomes.

- Observation only, with permission, and no personal data collected.

10. Future Scope

Future development may incorporate live waste-bin sensors, historical collection records, predictive risk analytics, geographical risk mapping and integration with relevant government waste-management reporting systems.

Nearer-term steps: repeat audits across more sites, add trend charts across several audit rounds, and add control charts once enough data exists.

11. How This Addresses the Evaluation Criteria

Criterion	Weight	How the project responds
Technical Implementation	25%	Tested calculation engine, validation, Occurrence computed from data, scenario recalculation
Functionality & Execution	20%	End-to-end loop that works offline: input, scoring, ranking, action, before and after
Problem Relevance & Impact	20%	Specific user and failures, tied to the four-stream rule, measured baseline and follow-up
Innovation & Originality	20%	Evidence-driven FMEA that separates projected from observed results, with a severity flag and a sensitivity check
Presentation & Pitch	15%	A clear story: measure, rank, act, verify

12. Submission Checklist

1. Project title and selected domain
2. Problem statement and proposed solution
3. Working prototype or MVP
4. Source code repository (GitHub)
5. Project presentation (PPT)
6. Brief documentation: implementation, technology stack, future scope

7. Demonstration of the working solution

13. To Fill In Before Submission

- Site, user and permission
- Failure list (8 to 12) and their consequence units
- Final scales and bands, reviewed by [reviewer]
- Real pilot numbers and dates
- One verified fact with its source, and a check of the 2026 Rules wording
- Final project name
- Full list of AI tools, APIs and libraries used

Disclaimer: WasteRisk 360 is an educational decision-support prototype inspired by ISO 31000. It does not claim ISO certification, regulatory approval or validated prediction of actual environmental or health outcomes.